

TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

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VERSION **v. 1.2**

RELATED SUPPORT

- TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2

This document contains the following Sections

Key Project Information

SECTION A– Description of project

SECTION B– Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

SECTION C– Duration and crediting period

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Appendix 2– Contact information of Project participants (mandatory)

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KEY PROJECT INFORMATION

GS ID of Project	11190
Title of Project	Household biogas plants in selected districts of Assam
Time of First Submission Date	20/05/2021
Date of Design Certification	N/A
Version number of the PDD	3.2
Completion date of version	25/11/2022
Project Developer	Value Network Ventures Advisory Services Pte. Ltd.
Project Representative	Value Network Ventures Advisory Services Pte. Ltd.
Project Participants and any communities involved	Biogas Service Center, Value Network Ventures Advisory Services Pte. Ltd.
Host Country (ies)	India
Activity Requirements applied	☒ Community Services Activities ☒ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☒ Small Scale ☐ Large Scale
Other Requirements applied	Not Applicable
Methodology (ies) applied and version number	AMS-I.E, version 12
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular

☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6.)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emission reductions	16,572	tCO ₂ e
3	Improvement in health and decrease in illness	100% users are expected in improved health and decrease in illness due to replacement of firewood based cooking practice	Number of affected users
7	Access to affordable and clean energy services	4,547 households (users) are expected to access clean energy	Number of affected users

SECTION A. DESCRIPTION OF PROJECT

A.1 Purpose and general description of project

>>The project activity involves bundling 4,547 household biogas plants located in rural parts of Assam in India commissioned from 22 May 2020 onwards. The biogas plants are constructed based on deenbandhu model. The project biogas plant replaces commonly used inefficient wood fired mud stoves thereby replacing firewood consumption.

The household biogas plants utilize cattle dung in the digester for the production of biogas which is burnt in gas stoves for household cookings. This leads to reduction of greenhouse gas emissions by displacing conventionally used non-renewable biomass with renewable biogas. In addition, the hygienic conditions in the rural areas will be improved by an appropriate disposal of waste. Further, residue from the bio digesters can be used as organic fertilizer and will improve soil fertility in rural areas.

Project activity will contribute towards sustainable development by replacing firewood with biogas generated from the biodigesters.

Implementation and operational framework:

The household biogas systems are constructed by approved RETs. Biogas Service Center has entered into agreement with RETs for identification, construction and enrollment of the household into the project. By joining the project each biogas system owner transfers the carbon credits rights to Biogas Service Center in lieu of free maintenance, technical support and necessary parts during the lifetime of the project. Biogas service center through its network of RETs and supervisors ensure continuous operation of each project biogas system.

A.1.1. Eligibility of the project under Gold Standard

>> As per section 4.1.3 of GS4GG Principles & Requirements version 1.2 of October 2019 'A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard published Activity Requirements and/or Gold Standard Approved Methodologies associated with it or as referenced in Gold Standard Product Requirements'.

The project falls under GG4GG Community Services Activity Requirements as described below:

Criteria	Applicability justification
2.2.1 All CSA Projects shall lead to climate change mitigation and/or adaptation by providing or improving access to services/resources at the household or community or institution level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	The project lead to climate change mitigation by providing biogas system (access to service) at household level. The project will provide thermal energy and waste management.
3.3.1 Certain Impact Quantification methodologies allow projects to account Suppressed Demand scenario when establishing a baseline. In such cases, the application of Suppressed Demand baseline is limited to Small Scale and Microscale Projects. Where a Suppressed	The project does not demonstrate baseline with Suppress demand. Hence, this criteria is not applicable.

Demand baseline is applied, it is not possible to 'stack' Gold Standard Certified Impact Statements or Products as the definition of the baseline may be contradictory.	
3.3.4 Legal ownership: (a) Projects involving the distribution of a large number of devices for services such as heating, cooking, lighting, electricity generation, water treatment technology such as water filter, etc. shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with the FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. (b) The transfer of Product ownership shall be discussed during local stakeholder consultations for projects.	Implementation of the proposed project doesn't involve any activity that causes alteration of any resource; therefore, acquiring any specific legal right to do so is not applicable. However, the entitlement of the emission reductions generated by the project shall be transferred to the project developer from the beneficiary households through a signed covenant. The carbon credit rights transfer is transparently and willingly shared with project developer. All benefits to join in the project and carbon credit rights transfer and sharing were discussed in the local stakeholder consultation.

General eligibility criteria under Para 3 of GS Principles and Requirements:

Criteria	Applicability justification
Types of Project: Eligible projects shall include physical action/implementation on the ground. Pre-identified eligible project types are identified in the Eligibility Principles and Requirements section.	The project falls under 'Renewable energy" type: Waste management and handling: Management of animal waste (cattle dung) to deliver biogas at household level. The project aims 4,547 households with one 3 m³ system at each household.

Location of Project: Projects may be located in any part of the world.	The project is located in Assam state of India and hence meets the requirement.
Project Area, Project Boundary and Scale: The Project Area and Project Boundary shall be defined. Projects may be developed at any scale although certain rules, requirements and limitations may apply under specific Activity Requirements, Impact Quantification Methodologies and Products Requirements. In order to avoid double counting the Project shall not be included in any other voluntary or compliance standards programme unless approved by Gold Standard (for example through dual certification). Also, if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature, the project shall demonstrate that there is no double counting of impacts at design and performance certification (for example use of similar technology or practices through which the potential arises for double counting or misestimation of impacts amongst projects).	Project Area and Boundary is described under section A.4 below. Scale: The project falls under waste handling and disposal with emission reductions 16,572 tCO₂ with installed energy output of 1.59 MW_{thermal} (Below the threshold of 45 MW_{thermal}). Hence, the project falls under small scale projects. The project activity as a whole or any project digester is not part of any other registered carbon credit project (such as CDM, VCS etc.) project. This is confirmed through end user agreement and can be cross checked from publicly available sources such as UNFCCC, VCS etc.
Host Country Requirements: Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	The project activity is a voluntary initiative and does not require any legal compliance/approval from host country. However, the project technology is approved and successful as approved by host country in its national sectoral policy. Hence, the project meets the criteria.

Contact Details: As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details of all Project Participants; AND in case of an organisation (iii) the legal registration details and (iv) documentation by the governing jurisdiction that proves that the entity is in good standing (defined as being a legal or other appropriate entity registered in or allowed to operate within the required jurisdiction and with no evidence of insolvency or legal/criminal notices placed against it or any of its Directors). Gold Standard retains the right (at its own discretion) to refuse use of the Standard where reputational concerns are highlighted.	The project developer is VNV Advisory Services and contact details of project developer is provided in the PDD. The entity is legally registered in the host country to operate and no insolvency or legal/criminal notices placed against it or any of its Directors.
Legal Ownership: Full and uncontested legal ownership of any Products that are generated under Gold Standard Certification, (for example carbon credits) shall be demonstrated. Where such ownership is transferred from project beneficiaries this must be demonstrated transparently and with full, prior and informed consent (FPIC). Note that for certain Project types there is a requirement for full and uncontested legal land title/tenure to be demonstrated. These are contained within specific Activity or Product Requirements. All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	Implementation of the proposed project doesn't involve any activity that causes alteration of any resource; therefore, acquiring any specific legal right to do so is not applicable. However, the entitlement of the emission reductions generated by the project shall be transferred to the project developer from the beneficiary households through a signed covenant.
Other Rights: As well as legal title and ownership, the Project Developer shall also demonstrate where required uncontested legal rights and/or permissions concerning changes in	Implementation of the proposed project doesn't involve any activity that causes alteration of any resource; therefore, acquiring any

use of other resources required to service the Project (for example, access rights, water rights etc.). Any known disputes or contested rights must be declared immediately to Gold Standard by the Project Developer and resolved prior to further project implementation in affected areas	specific legal right to do so is not applicable. However, the project technology is approved and successful as approved by host country in its national sectoral policy. Hence, the project meets the criteria.
Official Development Assistance (ODA) Declaration: All Project Developers applying for project activities located in a country named by the OECD Development Assistance Committee's ODA recipient list and seeking Gold Standard Certification for carbon credits shall declare the Official Development Assistance (ODA) support. The Project Developer shall follow the GHG Emissions Reduction & Sequestration Product Requirements and submit the declaration at the time of Design Certification.	No Official Development Assistance involves in the project activity. The cost of bio-digesters are born by users. Project developer has submitted ODA declaration.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

>>Implementation of the proposed project doesn't involve any activity that causes alteration of any resource; therefore, acquiring any specific legal right to do so is not applicable. However, the entitlement of the emission reductions generated by the project shall be transferred to the project developer from the beneficiary households through a signed covenant.

A.2 Location of project

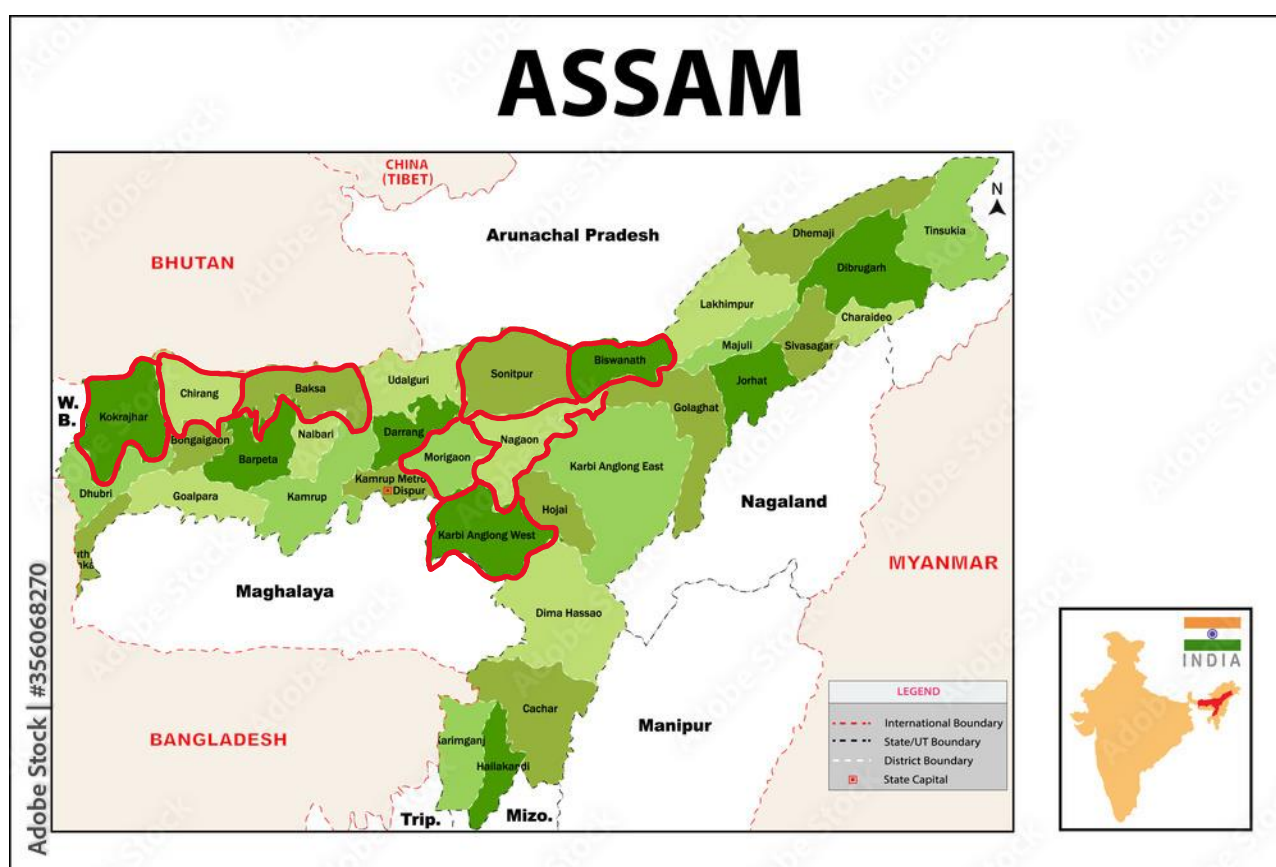
>> Host country: India

State/Region/Province: Assam

City/Town/community etc.:

Districts	Geo-coordinates
West Karbi Anglong	25° 51' 0" N, 92° 32' 0" E
Baksa	26° 34' 51" N, 91° 25' 13" E

Nagaon	26°35'038" N, 92°67'95" E
Morigaon	26°25'33" N, 92°34'24" E
Sonitpur	26° 37' 48" N, 92° 48' 0" E
Biswanath	92° 16'E, 26° 30'N
Chirang	26° 34' 48" N, 90° 36' 36" E
Kokrajhar	26° 24' 0" N, 90° 16' 12" E

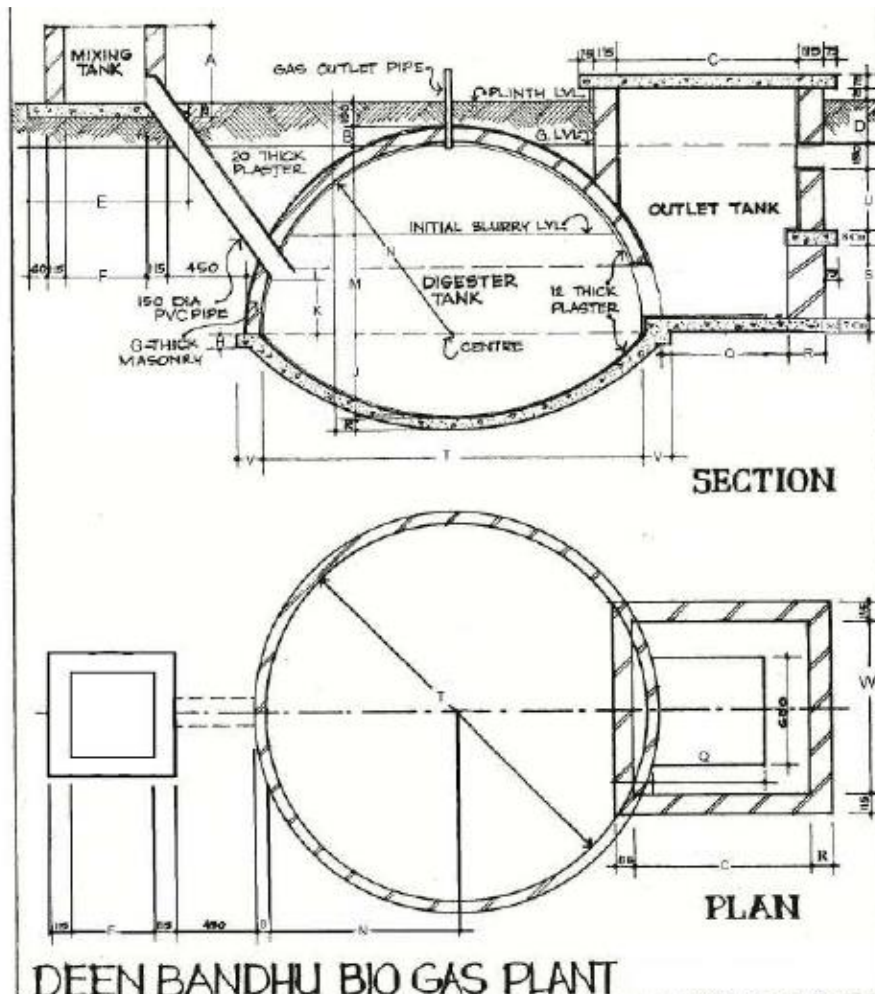


A.3 Technologies and/or measures

>> The household bio-digesters involved in the project activity are of Fixed –dome Digester technology (Deenbandhu model) of 3m³ size. The major feed cattle dung is mixed with water and fed into the plant through the inlet chamber of the plant. This waste is converted into biogas with the help of a special type of anaerobic bacteria. The digested material, which comes out of the plant, is enriched manure.

Deenbandhu Model (Fixed Dome):

The main feature of a Deenbandhu biogas plant is the fixed underground digester chamber, constructed with a layer of bricks and an additional layer of cement mortar forming the roof above. Connected to the underground chamber is an inlet tank, through which manure is fed into the plant. The manure then ferments separating the slurry from the methane gas which rises and collects at the top of the digester tank, and is released through the gas outlet pipe. The slurry passes into the outlet tank where it is ejected from the plant and can be used as fertilizer on the field.



The Deenbandhu biogas model is approved by MNRE (Ministry of New and Renewable Energy, Govt. of India) and constructed as per Indian Standard. The average life of brick masonry constructed biogas plants are around 20 years as per MNRE¹. Although it depends on the operational practice. In order to ensure continuous operation, users are provided with free maintenance service by Biogas service center as part of agreement for which records to be maintained.

A 3m³ biogas system can produce 3m³ of biogas in a day (24 hours). The ratio of cattle dung and water is 1:1 to feed in the digester. Since, the household system is meant for daily consumption and feeding, there is no storage is required or designed. However, the digester tank can hold the equivalent capacity of biogas. Biogas slurry is free from weed seeds, foul smell and pathogen. It contains major plant nutrients which nourish the soil to accelerate the growth of plants, especially for root growth which enhances crop yield in a sustainable manner. It enhances the aeration and water holding capacity of soil for root penetration resulting in better growth.

A.4 Scale of the project

>>The project falls under waste handling and disposal with emission reductions of 16,572 tCO₂ per annum with installed energy output of 1.59 MW_{thermal} (below the threshold of 45 MW_{thermal}). Hence, the project falls under small scale projects.

A.5 Funding sources of project

>> No public funding from parties included in Annex I to the UNFCCC, is available to the project. No Official Development Assistance involves in the project activity. The cost of bio-digesters are born by users.

¹ https://www.ctae.ac.in/images/editorFiles/file/CTAE-2017/Biogas%20FAQ_English.pdf

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

>> Methodology : AMS-I.E - Switch from non-renewable biomass for thermal applications by the user

Version : Version 12.0

Reference:

<https://cdm.unfccc.int/methodologies/DB/BLVN9ULD1FRUVS2LYWW6WPYN9W78E>

B.2. Applicability of methodology (ies)

>>

Conditions	Applicability
This methodology comprises of activities to displace the use of non-renewable biomass by introducing renewable energy technologies to households, communities, and/or institutions such as schools, prisons or hospitals (hereinafter referred as end-users). Examples of these technologies include, but are not limited to: (a) Cookstoves using renewable biomass, such as briquettes, pellets, and woodchips; (b) Biogas stoves; (c) Bio-ethanol stoves; (d) Electric cookstoves, including induction cookstoves (hereafter electric cookstoves) that receive electricity entirely from: (i) An integrated renewable energy generating device, grid or mini-grid	The project aims to displace non-renewable biomass by using cattle-dung based biogas system at household level. Hence, the condition is met by the project.

that is 100 per cent powered by renewable energy sources; or (ii) A renewable energy system that is also connected to the grid via net-metering, where, on an annual basis, the electricity generated by the renewable energy system is larger than the annual electricity consumed by the electric cookstoves and any other loads (e.g. lighting, fans, TV) connected.	
Project participants are able to show that non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	The forest cover of India has decimated from nearly 40% of India's geographical area a century ago to 22% in 1951 and to 20.55% in 2001². This indicates that large-scale deforestation is prevalent since 1989. Furthermore, it may be noted as per data (tables T1 and T11) from the Global Forest Resources Assessment 2010 (FRA2010) that the wood removals (Industrial round wood removals + Wood fuel removals) have increased 23.56% from (35,055,000+ 213,169,000 = 248,224,000) cubic meters in 1990 to (45,957,000+260,752,000=306,709,000) cubic meters in 2005. State of Forest Report (FSI) report in 1987 i.e., prior to 1989 clearly states for India that the firewood consumption in 1987 is estimated at 157 million tonnes or 235 million cu.m³.

² State of Forest Report 2001

³ <http://www.fsi.nic.in> (page 46; section 3.7)

	However, the production of firewood from forests estimated by FSI (Forest Survey of India) is only 40 million cu.m⁴. Thus, there was a gap of 195 million cu.m. in demand and production of firewood. Thus it is established that non-renewable biomass has been used in India since 31 December 1989, using official reports of the Government of India.
In the case that technologies using renewable biomass are used under the project activity, this methodology is applicable where all emissions related to processing of biomass are fully accounted for and biomass is sourced from biomass residues and/or a dedicated plantation of the CDM project activity, meeting the following conditions: (i) For projects that use biomass residues, prior to the implementation of the project activity, the biomass residues have not been collected and used but been left for decay and would, in the absence of the project activity, continue to be left for decay; and	Biomass is not used in the project activity. Hence, the conditions are not applicable.

⁴ <http://www.fsi.nic.in> (page 46; section 3.8)

(ii) For projects that use biomass residues from a production process (e.g. production of sugar or wood panel boards), the implementation of the project does not result in an increase of the processing capacity of raw input (e.g. sugar, rice, logs, etc.) or in other substantial changes (e.g. product change) in this process; and (iii) The biomass used by the project facility is not stored for more than one year; and (iv) In the case biomass from dedicated plantations are used, the applicability conditions of TOOL16 "Project and leakage emissions from biomass" are satisfied.	
For electric cookstoves with integrated renewable energy device or with grid connected renewable energy system employing net metering, project participants shall demonstrate that, on an annual basis, at least 80% of the electricity generated is consumed by the electric	The project does not involve any electric cookstove and hence the condition is not applicable for the project.

cook stoves (i.e. 20% or less of electricity is consumed by other loads connected).	
For electric cook stoves, in all cases under paragraph 2(d) above where back-up diesel generators are used, this methodology is only applicable when no more than 1% of total electricity supply occurs from back up diesel generators on an annual basis.	The project does not involve any electric cookstove and hence the condition is not applicable for the project.
Under this methodology, emission reductions cannot be claimed only due to fuel-switch aspect and proposed project activities shall introduce new renewable energy based technologies, i.e. technology switch is also involved.	The project involves technology switch from conventional cooking system to biogas based cooking system. Hence, applicability condition is justified.
The methodology is applicable for technologies displacing use of non-renewable biomass by renewable energy	Project activity involves installation of biodigesters and biogas thus produced will displace the use of non renewable biomass to major extent. Therefore, condition is justified.
Project participants shall describe in the PDD/PoA-DD the proposed method for distribution of project devices and how the double counting of emission reductions has been addressed, for example, using methods such as unique identifications of product and end-user locations (e.g. programme logo), to prevent double counting of emission.	As in when a digester is commissioned and agreed for joining the project, a unique id is allotted and end user agreement is executed to avoid any double counting.

For project activities introducing bio-ethanol cookstoves, project participants shall demonstrate that the bioethanol cookstoves are designed, constructed and operated to the requirements (e.g. with regard to safety) of a relevant national or local standard or comparable literature. Latest guidelines issued by a relevant national authority or an international organisation may also be used.	The project activity does not involve any bio-ethanol cookstoves. Hence, applicability condition is not applicable.
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B.3. Project boundary

>>The project boundary is the physical, geographical site of the use of biomass or the renewable energy. Therefore, the project boundary encompasses the sum of all the 4,547 physical geographical sites of all individual biogas plants (digester system, pipe leading to the stove and the stove itself) realized by the project activity.

Source		GHGs	Included?	Justification/Explanation
Baseline	Firewood	CO ₂	Yes	The major source of emissions in the baseline due to burning of firewood.
		CH ₄	No	Excluded for simplification, this is conservative
		N ₂ O	No	Not applicable for the project activity
Project	Firewood	CO ₂	Yes	Project emissions from continued use of fire-wood shall be accounted
		CH ₄	No	Not applicable for the project activity
		N ₂ O	No	Not applicable for the project activity

B.4. Establishment and description of baseline scenario

>> As per "AMS I.E- Switch from non-renewable biomass for thermal applications by the user, Version 12 'the baseline scenario would be the use of fossil fuels for meeting similar thermal energy needs'.

The project activity involves the installation of anaerobic biodigesters for the production of biogas which replaces non-renewable biomass, used as a fuel for household cooking purposes. A baseline survey has been done to estimate the average firewood consumption prior to project activity at user point following sampling standard as per UNFCCC 'Sampling and surveys for CDM project activities and programmes of activities' version 07 and guideline 'Sampling and surveys for CDM project activities and programmes of activities' version 04.

A simple random sampling approach has been applied as the target population are of homogeneous nature. In common practice fire-wood is used for household cooking purposes in India. In Assam 72% of rural population depends on fire-wood for cooking purpose. This is also established as part of the project specific survey. All surveyed users used firewood as fuel for domestic needs. Therefore, fire-wood in absence of the project biogas is the baseline scenario. Other fuel options like LPG and coal are not economic compared to easily available firewood. Cattle dung is not a suitable option compared to firewood, as dung is used as manure in cultivation fields and to use as fuel it needs specific preparation which needs time, drying place etc. Therefore, the most suitable option is firewood. The project targets only those users who uses firewood as fuel.

The baseline survey was done by a third party covering various districts of Assam in August 2018. Below details are provided related to baseline survey:

Survey design: The household's survey was done to identify the different parameters of interest. Kitchen Performance Test (KPT) was done to identify the baseline fuel-wood consumption.

Sampling Design: The survey was implemented with the objective to assess the fuel-wood consumption and number of castles specifically to know the feasibility for the biogas technology promotion. GS Methodology "Technologies and Practices to Displace Decentralized Thermal Energy Consumption" was followed for minimum sample size.

COV	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
90/10 precision	12	26	45	70	101	137	179	226	279

For the survey, the desired 90/10 confidence/precision has been taken. For this, sample to be taken for the survey is 101 samples as per table above. Sample is also calculated using Guidelines for Sampling and Survey for CDM project Activities and Program of Activities". The sample calculation for the mean value parameters for 90/10 confidence/precision is as follows:

$$n \geq \frac{1.645^2 NV}{(N-1) \times 0.1^2 + 1.645^2 V}$$

The target population considered 13000 households in rural areas of Assam in different districts. Simple random sampling has been followed as the nature of rural households in Assam is almost similar. Considering the Coefficient of Variance as 0.6 and the number of population as 13000 households, the number of samples required is calculated as n=97. Since the table above provides the number of sample as 101, the 101 households are taken as sample households for KPT and survey conservatively. Out of the households identified using traditional cook-stoves in rural areas of Assam, 101 samples were randomly taken. A total of 101 households were visited for the survey in different 22 districts taken randomly from the list of potential 13000 households. A random function was used to select the sample randomly.

Results:

Baseline Fuel type:

Sl.No.	Types of fuel	Response	
		Number	Percentage
1	Firewood	101	100%
2	Agri-residue	0	0
3	LPG	0	0
4	Others	0	0

Fuel consumption kg/day/household:

Sl.No.	Days	Quantity in kg	Standard Deviation
1	Day-1	11.42	0.94
2	Day-2	11.65	0.91
3	Day-3	11.53	0.80
	Total (Mean)	11.53	0.91

B.5. Demonstration of additionality

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	The project falls under GG4GG Community Services Activity Requirements. As per Annex-B Positive list under 'GG4GG Community Services Activity Requirements' the project meets the criteria 3 'Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in ≤ 600 MWh of energy savings per year or ≤ 600 tonnes of emission reductions per year'.
Describe how the proposed project meets the criteria for deemed additionality.	Considering a biogas plant can save around 4.20 tonne/year of firewood which is used for cooking in baseline, the emission reduction as per applied methodology (AMS-I.E, version 11) is calculated as below: $B_y = N_{HH} \times (BC_{BL,HH,y} - BC_{PJ,HH,y})$ Where, B_y = Quantity of woody biomass that is substituted or displaced in tonnes N_{HH} = Number of households in the project activity, number. It is considered 1 considering a single unit. $BC_{BL,HH,y}$ = Average annual consumption of woody biomass per household before the start of the project activity, tonnes/household/year 4.2 tonnes/HH/year is considered. $BC_{PJ,HH,y}$ = If it is found that pre-project devices were not completely displaced but continue to be used to some extent, average annual consumption of woody biomass per household in the pre-project devices during the project activity, tonnes/household/year. It is considered conservatively that entire firewood is replaced.

	$B_y = 1 * (4.20 - 0)$ $= 4.20 \text{ tonnes}$ Therefore, emission reduction is $BE_y = B_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossil_fuel}$ $= 4.2 * 100\% \text{ (most conservative value)} * 0.0156 * 64.4$ $= 4.22 \text{ tCO}_2\text{per year}$ Therefore, with conservative assumptions, the emission reduction for an individual unit is far below 600 tonnes of emission reduction per year. Hence, the proposed project activity falls under the 'Positive List' and hence under Principle 5 – Financial Additionality & Ongoing Financial Need, the project is considered deemed additional.
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B.5.1 Prior Consideration

>>As per GS4GG rule for retroactive projects, project documents need to submit to GS within one year of the project start date to meet prior consideration. Only biogas plants commissioned from one year prior to GS submission shall be considered. The project start date is 22/05/2020 and initial documents were submitted to GS on 20/05/2021. Therefore, prior consideration is met for the project activity.

B.5.2 Ongoing Financial Need

>> Ongoing Financial Need shall be demonstrated at Design Certification Renewal (Refer clause 3.5.2.2 of GS4GG 'principle and requirements').

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)

13-Climate Action (mandatory)	SDG target (13.2)- Integrate climate change measures into national policies, strategies and planning	Amount of emission reductions achieved from the project activity.
7-Affordable and clean energy	SGD targets: 7.1- By 2030, ensure universal access to affordable, reliable and modern energy services 7.2- By 2030, increase substantially the share of renewable energy in the global energy mix	7.1.2-Proportion of population with primary reliance on clean fuels and technology
3-Good health and well being	SDG target 3.9- By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination.	3.9.1 -Mortality rate attributed to household and ambient air pollution.

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact >> SDG 13:

As per "AMS I.E- Switch from non-renewable biomass for thermal applications by the user, Version 12, the baseline emissions (BE_y) are calculated as:

$$BE_y = B_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossil_fuel} \dots\dots\dots (1)$$

Where,

BE_y = Baseline emissions during the year y in t CO₂e

B_y = Quantity of woody biomass that is substituted or displaced in tonnes

f_{NRB,y} = Fraction of woody biomass used in the absence of the project activity in year y that can be established as non-renewable biomass (f_{NRB})

NCV_{biomass} = Net calorific value of the non-renewable woody biomass that is substituted (IPCC default for wood fuel, 0.0156 TJ/tonne)

EF_{projected_fossilfuel} = Emission factor for the substitution of non-renewable woody biomass by similar consumers. Use a value of 64.4 tCO₂/TJ for South Asia.

B_y is determined by using option (a) paragraph 29 (a) of the methodology as follows:

'Calculated as the product of the number of households multiplied by the estimate of average annual consumption of woody biomass per household that is displaced by the project activity (tonnes/household/year)';

$$B_y = N_{HH} \times (BC_{BL,HH,y} - BC_{PJ,HH,y}) \dots\dots\dots (2)$$

Where,

N_{HH} = Number of households in the project activity, number

$BC_{BL,HH,y}$ = Average annual consumption of woody biomass per household before the start of the project activity, tonnes/household/year

$BC_{PJ,HH,y}$ = If it is found that pre-project devices were not completely displaced but continue to be used to some extent, average annual consumption of woody biomass per household in the pre-project devices during the project activity, tonnes/household/year

$BC_{BL,HH,y}$ has been determined as per third party survey for the sample of households explained in section B.4 above.

Fraction of woody biomass used in the absence of the project activity in year y that can be established as non-renewable biomass ($f_{NRB,y}$) is determined as per methodological tool 'Calculation of the fraction of non-renewable biomass' version 03 as follows:

The fraction of woody biomass that can be established as non-renewable, is:

$$f_{NRB} = \frac{NRB}{NRB + RB} \dots\dots\dots (3)$$

Where:

f_{NRB} = Fraction of non-renewable biomass (fraction or %)

NRB = Quantity of non-renewable biomass (t/yr)

RB = Quantity of renewable biomass (t/yr)

The quantity of non-renewable biomass consumed in the applicable area (NRB) shall be determined as the difference between the total consumption of woody biomass in the applicable area (H) and the quantity of renewable biomass that can be sustainably harvested in the applicable area (RB):

$$NRB = H - R \dots\dots\dots (4)$$

Where: H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes).

The total consumption of woody biomass (H) is calculated using the following equation, accounting for all consumption within the applicable area (not only wood fuel but also timber and industrial consumption):

$$H = HW \times N + CE + N \dots\dots\dots (5)$$

Where:

HW = Average consumption of wood fuel per household, including fuelwood and charcoal, in the applicable area in the relevant period (tonnes//household)

CE = Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

NE = Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

N = Number of households consuming wood fuel within the applicable area in the relevant period (number)

Following paragraph 15 of the tool 'For the parameters HW and N, an aggregated value ($HW \times N$) may be used in the calculation, if official statistics provide total household wood fuel consumption ($HW \times N$)', Forest Survey of India report 2011, Annual fuelwood consumption in household sector and consumption of wood in House construction, Furniture and Agriculture is given below:

States	Assam
Annual woodfuel consumption in household sector (Tonne/year)	11421000
Non-domestic wood consumption (tonne/year)	8670250
Total wood consumption (tonne/year)	20091250

Procedure to estimate RB:

Renewable biomass (RB) in the country/region/area is estimated using the equation below:

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other})) \dots\dots\dots (6)$$

Where:

$MAI_{forest,i}$ = Mean Annual Increment of woody biomass growth per hectare in subcategory i of forest areas (t/ha/yr). For this parameter, publicly available research paper is used.

$F_{forest,i}$ = Extent of forest in sub-category i (ha). India state of forest report, 2021 has been referred for this parameter.

$MAI_{other,i}$ = Mean Annual Increment of woody biomass growth per hectare in subcategoryi of other wooded land areas (t/ha/yr). For this parameter, same value as for $MAI_{forest,i}$ is used.

$F_{other,i}$ = Extent of other wooded land in sub-category i (ha). India state of forest report, 2021 has been referred for this parameter.

P_{forest} = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (ha). India state of forest report, 2021 has been referred for this parameter.

P_{other} = Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other wooded land areas (ha). This data is not available and hence '0' is used on conservative side.

i = Sub-category i of forest areas and other wooded land areas. This data is not available and hence not used.

State	Assam
$MAI_{forest,i}$	0.974
RB	1823578.32

Accordingly, the fNRB would be:

State	H	RB	NRB (H-RB)	fNRB
Assam	20091250	1823578.32	18267671.68	90.9%

Project Emissions (PEy):

As per applied methodology AMS-I.E, version 12, project emissions are accounted for below activities:

CO2 emissions from on-site consumption of fossil fuels due to the project activity

CO2 emissions from electricity consumption by the project activity

Methane emission from solid waste disposal or waste water

Project emissions related to cultivation of feedstock

Project emissions from transportation

The project activity does not involve any of the above activity and hence, project emissions for the project activity is not applicable.

Leakage Emissions (LEy):

Leakage emissions (related to the non-renewable woody biomass saved by the project activity shall be assessed based on ex post surveys of users and the areas from which this woody biomass is sourced (using 90/30 precision for a selection of samples). The following potential source of leakage shall be considered: The use/diversion of non-renewable woody biomass saved under the project activity by non-project households/users that previously used renewable energy sources. If this leakage assessment quantifies an increase in the use of non-renewable woody biomass used by the non-project households/users that is attributable to the project activity, then By is adjusted to account for the quantified leakage. Alternatively, By is multiplied by a net to gross adjustment factor of 0.95 to account for leakages, in which case surveys are not required.

PP has opted default option, and By shall be adjusted with adjustment factor of 0.95 to account leakage.

Emission reductions:

Emission reductions are to be estimated based on the equation below:

$$ERy = BEy - PEy - LEy \dots\dots\dots (7)$$

SDG 3: Sample survey to be applied to estimate impact of SDG 3.

SDG 7: Sample survey to be applied to estimate impact of SDG 7.

B.6.2 Data and parameters fixed ex ante

Copy the table for each piece of data and parameter; use headings to group parameter tables by SDG

SDG13

Data/parameter	f _{NRB,y}
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Unit	Fraction
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass
Source of data	The data source is used publicly available sources such as India State of Forest Resources 2011, India State of Forest Resources 2021, Published Paper: Phytomass carbon pool of trees and forests in India etc.
Value(s) applied	90.9%
Choice of data or Measurement methods and procedures	Methodological tool: Calculation of the fraction of non-renewable biomass 03
Purpose of data	Baseline emissions estimation
Additional comment	This value will be fixed for the crediting period and will be reviewed during renewal of crediting period

Data/parameter	$BC_{BL,HH,y}$
Unit	tonnes/household/year
Description	Average annual consumption of woody biomass per household before the start of the project activity
Source of data	Baseline survey conducted in August 2018
Value(s) applied	4.2
Choice of data or Measurement methods and procedures	The survey was conducted to know the firewood consumption pattern in line with the methodology in the potential households that uses traditional cookstoves (baseline technologies). Survey was conducted following UNFCCC sampling standard 'Sampling and surveys for CDM project activities and programmes of activities' version 07 and guideline 'Sampling and surveys for CDM project activities and programmes of activities' version 04. A 90% confidence

	interval and 10% error margin applied for sample calculation
Purpose of data	Baseline emissions estimation
Additional comment	

Data/parameter	N_{HH}
Unit	Number
Description	Number of households in the project activity in year y
Source of data	As per PP's project database
Value(s) applied	4,547
Choice of data or Measurement methods and procedures	Project database as per the commissioning reports and end user agreements
Purpose of data	Baseline emissions estimation
Additional comment	

Data/parameter	$NCV_{biomass}$
Unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted
Source of data	AMS-I.E, version 12
Value(s) applied	0.0156 TJ/tonne
Choice of data or Measurement methods and procedures	IPCC default for wood fuel
Purpose of data	Baseline emissions estimation

Additional comment	
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Data/parameter	$EF_{projected_fossilfuel}$
Unit	tCO ₂ /TJ
Description	Emission factor for the substitution of non-renewable woody biomass
Source of data	Default value as per AMS-I.E, version 12
Value(s) applied	64.4 tCO ₂ /TJ
Choice of data or Measurement methods and procedures	Default value as per AMS-I.E, version 12
Purpose of data	Baseline emissions estimation
Additional comment	

B.6.3 Ex ante estimation of SDG Impact

>> SDG 13:

Baseline Emissions:

$$BE_y = B_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossil_fuel}$$

Where,

$$B_y = N_{HH} \times (BC_{BL,HH,y} - BC_{PJ,HH,y})$$

	Total population (N _{HH})	Baseline firewood consumption (tonne/year/HH)-BC _{BL,HH,y}	By (ton/yr)
3 m3	4,547	4.20	19,097.40

BC_{Pj,HH,y} (if it is found that pre-project devices were not completely displaced but continue to be used to some extent, average annual consumption of woody biomass per household in the pre-project devices during the project activity, tonnes/household/year) is considered zero during design certification stage. The same shall be monitored ex-post certification.

The fraction of woody biomass that can be established as non-renewable, is:

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

Accordingly, the fNRB would be: 90.9%

Accordingly, BEy would be:

$$BE_y = B_y \times f_{NRB,y} \times NCV_{biomass} \times EF_{projected_fossil_fuel}$$

By	fNRB	NCV	EF	BEy (tCO2e)
Assam				
19,097.40	90.9%	0.0156	64.4	17,445

Project Emissions (PEy): Nil as explained in section B.6.2 above.

Leakage Emissions: By is multiplied by a net to gross adjustment factor of 0.95 to account for leakages. Therefore, Leakage would be 872 tCO2

$$\begin{aligned}
 \text{Emission reductions} &= BEy - PEy - LEy \\
 &= 17,445 - 0 - 872.23 \\
 &= 16,572 \text{ tCO}_2
 \end{aligned}$$

SDG 3:

Baseline estimation	Ex-ante estimation
Prone to illness due to smoke released from burning of fire-wood	Reduction in smoke related illness due to use of biogas instead of firewood. Since, biogas is smokeless; 100% users are expected to get rid of kitchen smoke related illness.

SDG 7:

Baseline estimation	Ex-ante estimation
All users used fire-wood based cooking system which is not clean energy.	100% users are accessed to clean energy sources. This shall be monitored. It is assumed that with 100% operational status all users are accessed to clean energy source.

B.6.4 Summary of ex ante estimates of each SDG Impact

Year	Baseline estimate	Project estimate (Leakage)	Net benefit
Year 1	17,445	872	16,572
Year 2	17,445	872	16,572
Year 2	17,445	872	16,572
Year 4	17,445	872	16,572
Year 5	17,445	872	16,572
Total	87,223	4,361	82,860
Total number of crediting years	5		
Annual average over the crediting period	17,445	872	16,572

B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

SDG 13

Data / Parameter	BC _{PJ,HH,y}
Unit	tonnes/household/year
Description	Average annual consumption of woody biomass per household in the pre-project devices during the project activity, if it is found that pre-project devices were not completely displaced but continue to be used to some extent
Source of data	Survey
Value(s) applied	To be monitored

Measurement methods and procedures	Monitoring shall consist of estimation of all project devices or a representative sample thereof, at least once every two years (biennial)
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	Third party survey to be conducted following standard sampling approach
Purpose of data	Baseline Emissions estimation
Additional comment	

Data / Parameter	$N_{i,y}$
Unit	Numbers
Description	Number of project devices of type i operational in year y
Source of data	Survey
Value(s) applied	To be monitored
Measurement methods and procedures	Monitoring consist of checking of representative sample, to ensure that bio-digesters operating. Calculated as $N_{HH} * P_y$, where N_{HH} is the number of biodigesters installed in the project and P_y is Proportion of Bio-digesters operational estimated based on the sample survey. Following AMS-I.E version 12 requirement i.e. ‘In the specific case of biogas project activities, monitoring shall be conducted as per the requirements stipulated for “Proportion of $Nk,0$ that remain operating at year (nk,y)” under “AMS-I.I.: Biogas/biomass thermal applications for households/small users”. Accordingly, Monitoring of operability of the biogas systems shall be conducted using the following method: - Census of users or survey of the users at randomly selected sample sites;
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	Third party survey to be conducted following standard sampling approach. The “Standard: Sampling and surveys for CDM project activities and programme of

	activities” shall be used for determining the sample size to achieve 90/10 (for annual monitoring) or 95/10 (for biennial monitoring) confidence/precision levels.
Purpose of data	Baseline Emissions estimation
Additional comment	Net-to-gross adjustment factor of 0.89 is applicable in cases where the operationality is determined based on questionnaire survey, i.e. when using option (a) above, to account for uncertainties.

Data / Parameter	Date of commissioning of project device of type <i>i</i>
Unit	Date
Description	Actual date of commissioning of the project device
Source of data	Internal records
Value(s) applied	-
Measurement methods and procedures	-
Monitoring frequency	Fixed and recorded at the time of commissioning/distribution
QA/QC procedures	-
Purpose of data	-
Additional comment	-

Data / Parameter	Date of commissioning of batch <i>j</i>
Unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch
Source of data	Internal records
Value(s) applied	-

Measurement methods and procedures	-
Monitoring frequency	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch
QA/QC procedures	-
Purpose of data	-
Additional comment	-

SDG 3:

Relevant SDG Indicator	1 – Good health and well being (SDG 3)- Specific indicator 3.9.1
Data / Parameter	Improvement in health and decrease in illness
Unit	Qualitative
Description	Indoor air quality at usage point
Source of data	Sampling survey/annual usage survey/monitoring survey
Value(s) applied	To be monitored
Measurement methods and procedures	Improvement in health and decrease in illness will be assessed through interview with end users due to project implementation. Users opinion on indoor air quality due to biogas usage shall be collected during monitoring survey. In addition, training to the operation and maintenance technicians and field supervisors to be provided to increase awareness in safe operation and handling emergency situations.
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	Third party survey to be conducted following standard sampling approach.Publicly available data may be referred.
Purpose of data	Sustainable development assessment
Additional comment	

SDG 7:

Relevant SDG Indicator	2 – Affordable and clean energy (SDG 7)- Specific indicator 7.1.2
Data / Parameter	Access to affordable and clean energy services
Unit	Numbers
Description	Number of biogas system operational under the project activity

Source of data	Biogas User Survey
Value(s) applied	To be monitored
Measurement methods and procedures	Sample survey to confirm if project biogas systems are operational. Operational status will confirm that the users are accessed to affordable and clean energy.
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	Required sample size shall be determined following UNFCCC sampling standard
Purpose of data	Sustainable development assessment
Additional comment	

B.7.2 Sampling plan

>>

Sampling Approach:

Objectives and reliability requirements: The objective of the sampling plan is to achieve unbiased and reliable estimates of the proportion or the mean value of the key variables over the crediting period. The methodology applied for the project (AMS.I.E version 12) requires the project proponent achieving 95 percent confidence interval and a 10 percent margin of error when biennial inspection is chosen and when annual inspection is chosen 90 per cent. The table below provides the monitoring parameters that will be monitored annually:

Parameter	Type	Description
N_{HH}	Proportional parameter	Number of households (biogas system) in the project activity in operational per year
$BC_{PJ,HH,y}$	Mean value parameter	Average annual consumption of woody biomass per household in the pre-project devices during the project activity, if it is found that pre-project devices were not completely displaced but continue to be used to some extent.

Target Population: The target population for different parameters discussed in the table above are given below:

- For the proportional parameter; the target population is the bio-digester users listed in the project database.

- For the mean value parameter; the target population is the total number of operational biogas digester for which the emission reductions will be accounted for the monitoring period in question.

Sampling frame: All the households with biogas digester within the project will be the sampling frame.

Sampling Method: A simple random sampling will be adopted for estimating the sample size for the monitoring surveys. Simple random sampling is suited to populations that are homogenous (EB 86 annex 04).

Sample Size: The calculation of the required sample size for each parameter will be calculated at 90/10 confidence/precision as required for the annual monitoring. The sample size is determined using the Guidelines for Sampling and Surveys for CDM Project activities and Programme of Activities Ver. V4.0 (EB86, Annex 4).

Proportional Paramater (Operational status)

The minimum sample size to determine number of biogas system in operation using the procedure outlined in para 12 of appendix 1, EB 86 Annex 4, Guidelines for Sampling and Surveys for CDM Project activites and Programme of Activities Ver. 4.0.

$$n \geq \frac{1.645^2 N \times p(1 - p)}{(N - 1) \times 0.1^2 \times p^2 + 1.645^2 p(1 - p)}$$

Where:

n= Sample size

N = Total number of biogas system of type i installed under the project

p = expected proportion

1.645 = represents the 90% confidence required

0.1 = represents the 10% relative precision

Mean Value parameters (average annual consumption of woody biomass after project activity)

The minimum sample size for the monitoring parameters is determined using the equation given in para 51 of appendix 1, EB 86 Annex 4, Guidelines for Sampling and Surveys for CDM Project activities and Programme of Activities Ver. 4.0.

$$n \geq \frac{1.645^2 NV}{(N - 1) \times 0.1^2 + 1.645^2 V}$$

Where,

$V = (SD/Mean)^2$

n = Sample Size

N = Total number of Population

SD = Standard deviation

Mean = Mean for the average annual woody biomass consumption by pre-project device during project activity

1.645 = Represent 90% confidence required

0.1 = Represent the 10% relative precision

Out of the sample calculated above, conservative number of sample will be taken with appropriate enlargement of samples.

Data:

Field Measurements:

Checking of a representative sample of biogas system within the project to ensure that they are still operating (NHH).

The selection of households will be done using rand function in excel to generate random number and taking households from first up to the number of sample required by arranging the random number in ascending or descending order.

Determine average annual consumption of woody biomass per household in the pre-project devices during the project activity, if it is found that pre-project devices were not completely displaced but continue to be used to some extent.

The survey will be conducted annually with the objective to target 10 percent precision and to achieve 90 percent confidence. Since, survey will be done only once in a year and value to be scaled up for the whole year, the survey shall capture the parameter in

such a way that the resulted value represents seasonal fluctuations (e.g. survey questionnaire will capture the wood fuel consumption in different seasons of the year).

Quality Assurance/Quality Control:

A survey questionnaire will be prepared to seek responses of operating status (yes or no) of biogas digesters within the project activity. Users will be asked to provide their estimates on consumption of woody biomass after project activity. The survey will be performed by the project developer appointing a third party. During the survey, in order to anticipate any low response rate and answers bias, 10% oversampling will be applied. PP will ensure a physical on-site visit for the survey unless emergency situations (such as pandemic, national emergency, restrictions by host country regulation etc.) force to conduct the survey remotely. Hard copy questionnaire or tablet app shall be used for survey. PP will ensure training of field personnel engaged for survey to ensure quality data.

Analysis:

The project developer will collect, compile and analyze the data to derive the number of bio-digesters within the project still operational and average firewood consumption at each bio-digester user (if any). The developer will prepare "monitoring report" based on the survey report.

Implementation:

The survey questionnaire will be prepared, pre-tested and field personnel will be trained in conducting the survey to ensure the quality of data collected and the survey will be carried out as per defined frequency covering required data information. The schedule for implementing the sampling effort shall be defined prior to the field activity.

B.7.3 Other elements of monitoring plan

>> In addition to the parameters listed in section B.7.1 above, the following shall be monitored:

- Number of new devices distributed under the project activity, identified by the type of devices and the date of commissioning. A project database shall be maintained.

- Data to unambiguously identify the recipient of the new devices distributed under the project activity (e.g. name, address, phone number). A project database shall be maintained.

The bio-digesters are constructed by approved RETs (Renewable Energy Technicians) and handed over to end users. Although once commissioned, the biogas systems function smoothly, it requires maintenance during the lifetime. For that purpose the project participant 'Biogas Service Center' monitor the operational status of project bio-digesters and provide maintenance when required. Biogas Service center will keep records to maintenance/service and VNV will annually check the operational status of systems with sample survey.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1 Start date of project

>> 22/05/2020

C.1.2 Expected operational lifetime of project

>>15 years

C.2. Crediting period of project

C.2.1 Start date of crediting period

>>22/05/2020 or two years prior to the date of project design certification whichever is later.

C.2.2 Total length of crediting period

>> The project falls under GS community service activity and hence eligible for total 15 years issuance with certification renewal every 5 years. So the length of crediting period will be 5 year* 3 times.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Not applicable	No safeguarding principle is adversely impacted by the project activity.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project respects the key gender issues and requirements of gender-sensitive design and implementation of the project. SDG#5 is one of the impact areas of the project. The project is aimed to replace conventional wood fired mud stoves technology with biogas primarily for cooking. This will result in reducing use of firewood consumption or replacement of complete firewood usage at user place. Primarily in rural areas of India, cooking activity at household level is managed by women. Therefore, women are more exposed to the indoor air pollution and the associated hazard. Women in rural areas in most cases are also responsible for taking care of their children specially infants who need mother's support most of the time are bound to accompany their mother in kitchen. This situation leads to enhanced exposure of the women and
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	children to kitchen smoke and associated health consequences. Since the project aims to replace the polluting traditional cooking stoves with biogas cooking system, the primary beneficiary would be the women and children. Furthermore, the project is focused to the economically disadvantaged group of people, which also justifies the dimension of social inclusion in the project design.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The Government of India reaffirms its commitment to work for the realization of constitutional guarantee of equality, social justice and non-discrimination on the basis of sex, caste, community, language and religion. Ministry of women & child development, Govt. of India has taken various measures for gender equality/socio-economic development/empowerment of women. Out of these, the project positively contributes towards the national mission for empowerment of women through improvement of health and attaining vision for empowerment of women under national policy for women 2016 (Women participation will be ensured in the efficient use and spreading the use of solar energy, biogas, smokeless chulas and other technological applications to have positive influence on their life styles and a long term impact on meeting sustainable development goals)

Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	N/A
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	N/A

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes.

Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1 Summary of stakeholder mitigation measures

>> The stakeholder consultation was conducted on 02 October 2021. The stakeholders were invited through a public notice published in local news paper and also through e-mail/letter. The notice was published inline with requirement of GS on 03/09/2021. Stakeholders appreciated the proponent's transparent way of conducting the business and unanimously confirmed that the contact details provided by the proponent is enough for now and the project design is good enough to proceed with. Although no comments/concerns were raised by stakeholders for which mitigation measures were proposed and taken for monitoring, below comments were taken up by PP in the project activity:

Comment	PP's explanation
Any employment generation at local level?	The biogas service center shall use local people to maintain and survey as per monitoring requirement.
How carbon credit benefits to be shared with biogas owners?	The project developer informed that in lieu of carbon credit free maintenance service shall be provided to owners for lifetime of the project. This has been agreed by participating households.

E.2 Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	Each project user is provided a service book with contact details of Biogas Service Center and responsible supervisor to report any input/grievance or complaint.
GS Contact (mandatory)	help@goldstandard.org
Other	Telephone no: DIR: +91 80 42429985 CELL: +91 9845017165 E-mail: sandeep@vnvadvisory.com

All above methods to be used for taking continuous grievance and the same shall be monitored and incorporated in the monitoring report.

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	India adopted 'The protection of human rights Act' 1993 and the project is bound to follow the rules and regulation of host country. In addition, India has ratified 'International Convention on the Elimination of All Forms of Racial Discrimination :1969' 'International Covenant on Civil and Political Rights :1976', 'International Covenant on Economic, Social and Cultural Rights :1976' 'Convention on the Elimination of All Forms of Discrimination against Women :1981'.	Not applicable

		Therefore, the project developer and the project do respect nationally and internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of non discrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and	No	1. The project happens in individual households. It does not involve any women workforce which may lead to sexual harassment. 2. Yes. The project involves construction of biogas digesters at households. Trained labours are used for the same. Local people are engaged for the same. No discrimination either in gender or any other form is followed to engage local people. 3. The project does not has any scope to apply gender strategy as such. Although the project positively contributes	Not Applicable.

recommendations of an Expert Stakeholder(s)		towards the national mission for empowerment of women through improvement of health and attaining vision for empowerment of women under national policy for women 2016	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	The project leads to safe working condition and improvement in health as it will replace firewood as fuel with biogas which is clean and safe	Not applicable
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project area covers households which does not have any structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture. Hence, not applicable	Not applicable
>>			
Principle 4.2 Forced Eviction and Displacement			

Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project activity takes place within individual households. Therefore, no relocation of people takes place.	Not Applicable
>>			
Principle 4.3 Land Tenure and Other Rights			
a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	The project area covers households which does not have any structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture. Hence, not applicable	Not Applicable
b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?			
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	No. The project involves household biogas digesters. Therefore, it does not involve any influence towards indigenous people	Not Applicable

>>			
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	The project benefits households with clean fuel (biogas). There is no corruption provision in the project activity	Not Applicable
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and	No	The project does not require labour force for implementation of the project. Trained technicians are involved in construction and operation and maintenance of plants. Therefore, no forced labour is involved in the project. No child labour is involved	Not Applicable

implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property			
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5. requires an Expert Stakeholder opinion) The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	The project has lifetime of 15 years and expected to run beyond the project certification period. With carbon finance it is further strengthen to operate 100% project systems with continuous maintenance	Not Applicable.
>>			
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	No. The project will replace firewood use with biogas. Hence, it will reduce greenhouse gas emissions over the Baseline Scenario	Not Applicable
>>			

Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	No. The project uses inhouse cattle dung only	Not Applicable
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	No. Water in huge quantity is not required for the project which can impact the ground water level or any seasonal flow. Water is needed to mix with cattle dung while feeding in digester	Not Applicable
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	No. Water in huge quantity is not required for the project which can impact the ground water level or any seasonal flow. Water is needed to mix	Not Applicable

b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?		with cattle dung while feeding in digester	
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	No. Since the project takes place in individual households, no net degradation in existing landscape function and services takes place	Not Applicable
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	No. The project happens at individual households. There is no activity which can affect adversely the natural system to cause earthquake, landslides, erosion, flooding, draught or other extreme climatic conditions	Not Applicable.
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms	No	Not applicable. The project does not involve any crop production or cultivation	Not Applicable

or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?			
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	No. The project does not release any pollutants to the environment. The bioproduct obtained from biogas plant is biogas slurry which does not cause any harm to environment. Biogas slurry coming out of the digester gets stored in open bed near to the digester and regularly removed to use in agricultural fields. Biogas slurry is free from weed seeds, foul smell and pathogen. It contains major plant nutrients which nourish the soil to accelerate the growth of plants, especially for root growth which enhances crop yield in a	Not Applicable
>>			

		sustainable manner. It enhances the aeration and water holding capacity of soil for root penetration resulting in better growth ⁵ .	
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	Not applicable. The project does not involve any production process	Not applicable
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	Not applicable. The project does not involve any crop production or cultivation	Not applicable
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No		Not applicable

⁵ https://www.ctae.ac.in/images/editorFiles/file/CTAE-2017/Biogas%20FAQ_English.pdf

>>		Not applicable. The project happens at individual households	
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	Not applicable	Not applicable
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	Not applicable	Not applicable
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	Not applicable	Not applicable
>>			
Principle 9.11 Endangered Species			

a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?	No	Not applicable	Not applicable
>>			

APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

Organization name	Value Network Ventures Advisory Services Pte. Ltd
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APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Not applicable.

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption